Growth applications

1

a

i Exponential growth ii 60%

b

i Exponential decay ii 80%

c

i Exponential growth ii 105%

d

i Exponential decay ii 22%

2 $155\,432$

3 0.70 AU

4

a 1000 aphids **b** 32 000 aphids **c** 250 aphids

5

a 8 million

b

i 7130 million ii 9286 million

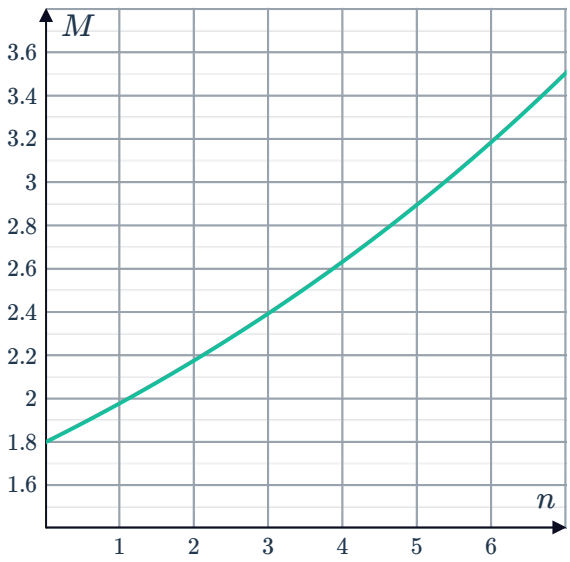
6

a $M = 2.4$

b

Months (n)	0	1	2	3	4	5	6
Mass (M)	1.8	2.0	2.2	2.4	2.6	2.9	3.2

c



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7

a

x	1	2	3	4
y	2	4	8	16

b 2

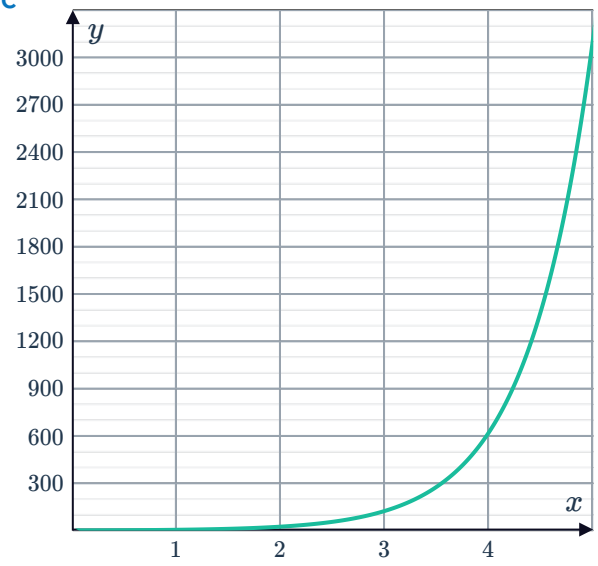
c $y = 2^x$

8

a No

b $y = (5)^x$

c



9

a 5

b $y = -6 \times 5^x$

10

a

No. of folds (f)	0	1	2	3
No. of rectangles (r)	1	2	4	8

b 2

c $r = 2^f$

d 2048

11

a $y = 10 \times 3^x$

b $y = 200 \times 1.25^x$

12

a $y = 750\,(1.03)^t$

b 869 rats

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13

a $\frac{5}{6}$

b $f(t) = \frac{5}{6}(1+1)^t$

c Growth

d 100%

14

a $A = 5100(1.112)^y$

b 11.2%

15

a $r = e^{0.09} - 1$

b $A = 2200(1 + 0.0942)^x$

c 9.42%

16

a

Seconds passed (t)	0	1	2	3	4	5
Number of bacteria (y)	1	2	4	8	16	32

b $y = 2^t$

c 2^{60}

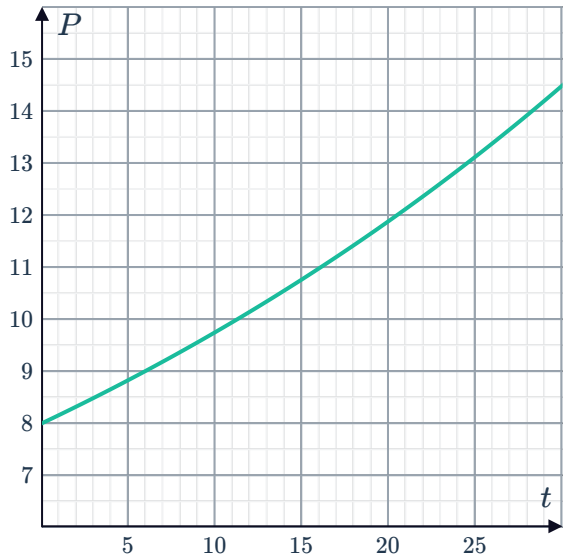
d $y = 2^{3600h}$

17

a

t	5	10	15	20	25	30
P	8.8	9.8	10.8	11.9	13.1	14.5

b



c 23 years

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18

a $y = 160\,(1.19)^t$

b $n = 9$

19

a $P = 90 + 5t$

b $Q = 90 \times (1.04)^t$

c The current company

d Opening his own business

20

a Treatment A

b 5

c Treatment A

d $72\,550$

21

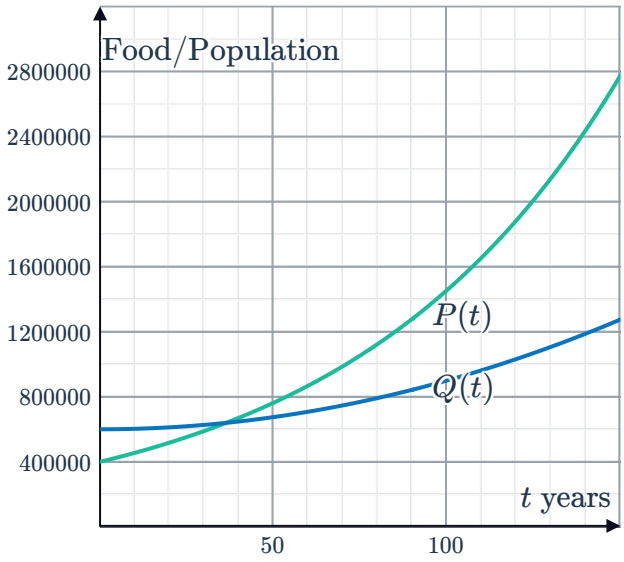
a $147\,850$

b There will be a surplus of food in 10 years time.

c $-555\,474$

d There will be a food shortage in 100 years time.

e



f 36.34 years

g No, the population increases at a faster rate than the food supply. Once the population has overtaken it will stay above and there will be an increasing gap.

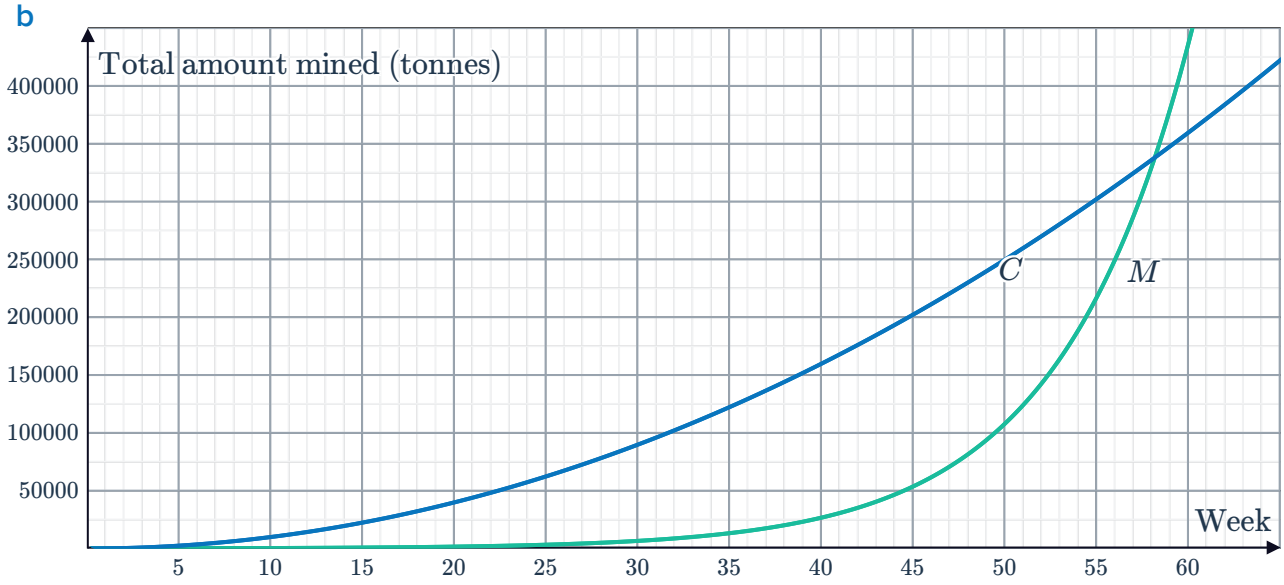
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22

a

n	M	C
1	100	100
5	175	352
10	708	2500
15	10 000	22 500

b



c Crest Corporation

d $(1, 100)$ and $(59.5, 353563.1)$, at the end of the first week of operations the two companies have mined the same total of 100 tonnes. The Crest Corporation then has a higher total amount mined until the approximately 59.5 weeks later when they are equal again at 353563.1 tonnes mined in total.

e Mint Corporation

Decay applications

23

a $y = 840\,(0.96)^m$

b 606 rodents

24

a $y = 2310\,(0.974)^n$

b 1328

25

a $y = 21\,(0.5)^n$

b 0.66 m

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26

a

i 179 ii 132

b

i 70% ii 40%

c If you have a high sugar diet, your blood sugar level peaks at higher amounts and decreases more rapidly, than for a person with a low sugar diet.

27

a Decreasing

b i No ii Yes iii Yes iv No

c The initial temperature of the substance

d $T = 145 \times 0.6^t$

e $0.88\,^\circ\text{C}$

f No

g Substance P

28

a $a = 5$ **b** $b = 4$ **c** $y = \dfrac{5}{16}$

29

a $y = 10 \times 3^x$ **b** $y = 200 \times 1.25^x$

30

a 2

b $f(t) = 2(1 - 0.625)^t$

c Decay

d 62.5%

31

a $A = 60(0.5)^{\frac{n}{8}}$ **b** 0.12 g

32

a $A = 300(0.5)^{\frac{n}{20}}$ **b** 0.59 g

33

a $A = 385(0.5)^{\frac{n}{5730}}$ **b** 321.11 g **c** $19\,823$ years

34

a $A = k \times (1 - 0.0245)^t$ **b** 2.45%

35

a $A = 8900(1 - 0.0769)^t$ **b** 7.69%

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36

a $A = 3400\,(1 - 0.2054)^t$

b 20.54%

37

a $P = (0.85)^t$

b Maximilian

c C

d 13%

e No

38

a 192.372 acres

b 185.03 acres

c $f\,(n) = 200\,(0.96186)^{n-1}$

d 3.814%

Financial applications

39

a $y = 2900\,(1.09)^t$

b $\$5778.43$

40 3313.46

41 241.70

42

a $\$\,(1.1)^t$

b $\$\left(1 + \dfrac{0.1}{12}\right)^{12t}$

43

a

i $A = 6100\,(1.083)^t$

ii $\$9088.08$

b

i $A = 6800\,(1.0025)^{12t}$

ii $\$7898.99$

c

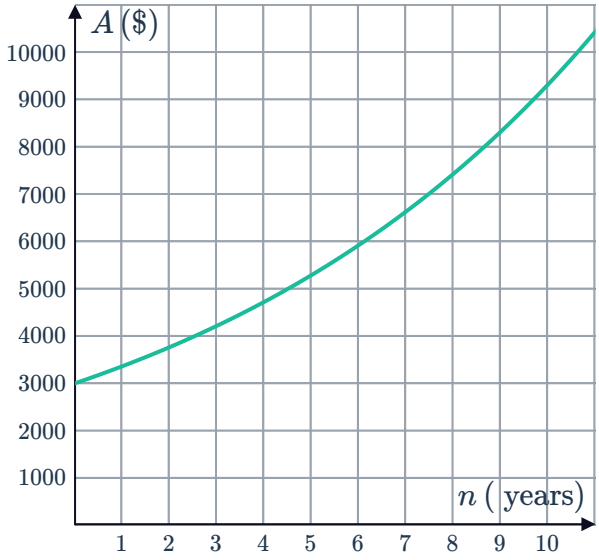
i $A = 3200\,(1.0013)^{4t}$

ii $\$4143.23$

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44

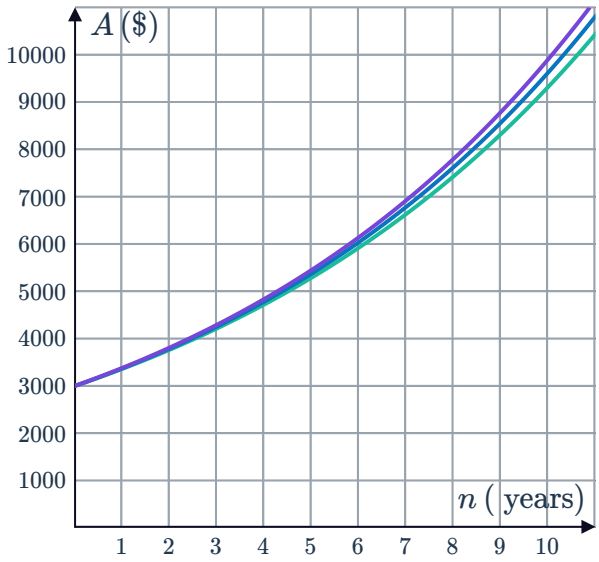
a



b $A = 3000 \times 1.06^{2n}$

c $A = 3000 \times 1.01^{12n}$

d



e $\$583.62$

45

a 0.02

b $A = 1000 \times 1.02^{n}$

c $\$1126.16$

46

a $M = 50\,000\,(1.032)^{t}$

b $P = 49\,000\,(1.061)^{t}$

c $\$9500.24$

d Elizabeth's

47

a Increasing by a factor of 2 each day.

b $\$64$

c $y = 4\left(2^{x-1}\right)$

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48

a

x	0	3	6	9
Company A's bill	200	218	236	254
Company B's bill	200	224.97	253.06	284.66

b Company B

49

a $f(n) = 125\,000 + 3000n$

b $g(n) = 20\,000\,(1.4)^n$

c

n	2	3	4
Increase in salary compared to previous year when working for a company	3000	3000	3000
Increase in salary compared to previous year when working for her own business	11 200	15 680	21 952

d Working for a company

e Starting her own business

50

a

i $A = P\,(1.053)^t$

ii $A = P\,(1.0125)^{4t}$

b 5.09%

c Bank A

51

a $A = 17\,000\,(1 + 0.0444)^{4n}$

b 4.44%

52

a

i $A = 600\,(1.0046)^{12t}$

ii $A = 600\,(1.048)^t$

b $600\,(1 + 0.0039)^{12t}$

c 0.39%

d Savings account

53

a $y = 322\,000\,(0.95)^m$

b $\$249\,157.46$

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54 **a** Sale price of property **b** 5.3%

c -5.3%

55 5.08%

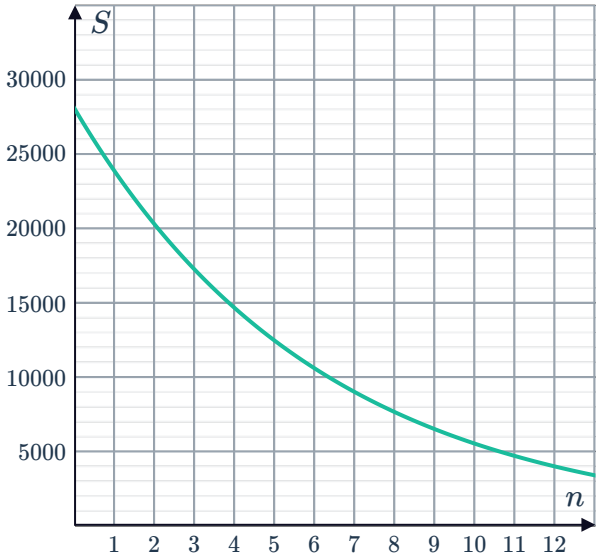
56 **a** $f(t) = 1.27\,(0.92)^t$ **b** 8%

c $\$0.47$

57 **a**

n	2	4	6	8	10	12
S	20 230	14 616.18	10 560.19	7629.73	5512.48	3982.77

b



c $\$17\,195.5$

d 9 years

58 **a** $A = 45\,000\,(0.78)^t$ **b** $\$16\,656.78$

59 **a** 0.01 **b** $A = 38\,200 \times 0.99^n$

c $\$35\,964.54$ **d** Motorbike

60 **a** $A = 25\,000\,(1 - 0.0031)^{52t}$ **b** 0.31%

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